Generalizable Label Distribution Learning

Supplementary Material

A.1 PROOF FOR THEOREM 1

THEOREM 1. *Let the KL divergence as the loss function bounded by a constant B [1], for any $\delta > 0$, with probability at least $1 - 2\delta$, the following bound holds for all $h \in \mathcal{H}$:*

$$R^{\mathcal{E}}(h) \leq \hat{R}^{\mathcal{E}}(h) + 4\sqrt{c}\mathcal{R}_{MN}(\mathcal{H}) + 4\sqrt{c}\mathcal{R}_{M}(\mathcal{H}) + 3B\sqrt{\frac{\ln(2/\delta)}{2MN}} + 3B\sqrt{\frac{\ln(2/\delta)}{2M}}, \tag{1}$$

where c is the number of labels and $\mathcal{R}(\mathcal{H})$ is the empirical Rademacher complexity of $\mathcal{H}$.

To prove Theorem 1, we start from the standard empirical Rademacher complexity bound on generalization error [2, 14]. Following [15], we have:

LEMMA 2. *Define $\mathcal{G} = \{(\boldsymbol{x}, \boldsymbol{d}) \mapsto \ell(\boldsymbol{d}, h(\boldsymbol{x})) \,|h \in \mathcal{H}\}$. For any $\delta > 0$, with probability at least $1 - \delta$, the following bound holds for all $\hat{h} \in \mathcal{H}$:*

$$R(h) \leq \hat{R}(h) + 2\mathcal{R}_n(\mathcal{G}) + 3B\sqrt{\frac{\log\frac{2}{\delta}}{2n}}, \tag{2}$$

where $\mathcal{R}_n(\mathcal{G})$ is the Rademacher complexity of $\mathcal{G}$ and n denotes the number of independent rademacher random variables.

Based on Lemma 2, we give a corollary which is a modification of the empirical Rademacher complexity bound on generalization error over training domains [11]:

COROLLARY 1. *For $\mathcal{P}^{(1)}, \cdots, \mathcal{P}^{(m)}$ independent samples from ε, and the KL divergence as loss function ℓ bounded by a constant B, for any $\delta > 0$, the following holds with confidence at least $1 - \delta$*

$$\frac{1}{M}\sum_{m=1}^{M} R_{\mathcal{P}^{(m)}}(h) \leq \frac{1}{M}\sum_{m=1}^{M} \hat{R}_{\mathcal{P}^{(m)}}(h) + 2\mathcal{R}_{MN}(\mathcal{G}) + 3B\sqrt{\frac{\ln(2/\delta)}{2MN}}, \tag{3}$$

where $\hat{R}_{\mathcal{P}^{(m)}}(h)$ is measured on $\mathcal{S}_{\mathcal{P}^{(m)}} = \left\{\left(\boldsymbol{x}_i^{(m)}, \boldsymbol{d}_i^{(m)}\right)\right\}_{i=1}^{N}$, a collection of N i.i.d. samples from $\mathcal{P}^{(m)}$.

With Corollary 1, we give a proof for Theorem 1.

PROOF. Corollary 1 tell us that with confidence at least $1 - \delta$,

$$\frac{1}{M}\sum_{m=1}^{M} R_{\mathcal{P}^{(m)}}(h) \leq \hat{R}^{\mathcal{E}}(h) + 2\mathcal{R}_{MN}(\mathcal{G}) + 3B\sqrt{\frac{\ln(2/\delta)}{2MN}}. \tag{4}$$

Thus, our goal is to provide an upper bound on

$$R^{\mathcal{E}}(h) - \frac{1}{M}\sum_{m=1}^{M} R_{\mathcal{P}^{(m)}}(h) \tag{5}$$

that holds uniformly for all $h \in \mathcal{H}$. Define

$$\Phi(P) = \sup_{h\in\mathcal{H}} R^{\mathcal{E}}(h) - \frac{1}{M}\sum_{m=1}^{M} R_{\mathcal{P}^{(m)}}(h). \tag{6}$$

If we construct P' by replacing any $\mathcal{P}^{(m)} \in P$ with $\mathcal{P}^{(m)\prime} \in \mathcal{E}$, then we have $|\Phi(P) - \Phi(P')| \leq \frac{B}{M}$. According to McDiarmid's inequality, with confidence at least $1 - \frac{\delta}{2}$

$$\Phi(P) \leq \mathbb{E}_{\mathcal{P}\sim\mathcal{E}}[\Phi(P)] + B\sqrt{\frac{\ln(2/\delta)}{2M}}. \tag{7}$$

[1] For KL divergence, 0 is replaced by a small constant γ [3, 15], resulting in there exists a constant $B \geq -\ln\gamma$ such that $\mathsf{KL}(\cdot,\cdot) \leq B$ (e.g. B=35 for $\gamma = 1 \times 10^{-15}$).

We proceed by bounding the expected value of $\Phi(P)$,

$$
\begin{aligned}
&\mathbb{E}_{\mathcal{P}\sim\mathcal{E}}\left[\sup_{h\in\mathcal{H}}\left(\mathbb{E}_{\tilde{\mathcal{P}}\sim\mathcal{E}}\left[R_{\tilde{\mathcal{P}}}(h)\right]-\frac{1}{M}\sum_{m=1}^{M}R_{\mathcal{P}^{(m)}}(h)\right)\right]\\
=&\mathbb{E}_{\mathcal{P}\sim\mathcal{E}}\left[\sup_{h\in\mathcal{H}}\mathbb{E}_{\tilde{\mathcal{P}}\sim\mathcal{E}}\left[\frac{1}{M}\sum_{m=1}^{M}\left(L_{\tilde{\mathcal{P}}^{(m)}}(h)-L_{\mathcal{P}^{(m)}}(h)\right)\right]\right]\\
\leq&\mathbb{E}_{\mathcal{P},\tilde{\mathcal{P}}\sim\mathcal{E}}\left[\sup_{h\in\mathcal{H}}\frac{1}{M}\sum_{m=1}^{M}\left(L_{\tilde{\mathcal{P}}^{(m)}}(h)-L_{\mathcal{P}^{(m)}}(h)\right)\right]\\
=&\mathbb{E}_{\mathcal{P},\tilde{\mathcal{P}}\sim\mathcal{E}}\mathbb{E}_{\sigma}\left[\sup_{h\in\mathcal{H}}\frac{1}{M}\sum_{m=1}^{M}\sigma^{(m)}\left(L_{\tilde{\mathcal{P}}^{(m)}}(f)-L_{\mathcal{P}^{(m)}}(f)\right)\right]\\
\leq&2\mathbb{E}_{\mathcal{P}\sim\mathcal{E}}\mathbb{E}_{\sigma}\left[\sup_{h\in\mathcal{H}}\frac{1}{M}\sum_{m=1}^{M}\sigma^{(m)}R_{\mathcal{P}^{(m)}}(f)\right]\\
=&2\mathbb{E}_{\mathcal{P}\sim\mathcal{E}}\mathbb{E}_{\sigma}\left[\sup_{h\in\mathcal{H}}\frac{1}{M}\sum_{m=1}^{M}\sigma^{(m)}\mathbb{E}_{(\boldsymbol{x},\boldsymbol{d})\sim\mathcal{P}^{(m)}}[\ell(h(\boldsymbol{x}),\boldsymbol{d})]\right]\\
\leq&2\mathbb{E}_{\mathcal{P}\sim\mathcal{E}}\mathbb{E}_{(\boldsymbol{x},\boldsymbol{d})\sim\mathcal{P}^{(m)}}\mathbb{E}_{\sigma}\left[\sup_{h\in\mathcal{H}}\frac{1}{M}\sum_{m=1}^{M}\sigma^{(m)}\ell\left(h\left(\boldsymbol{x}\right),\boldsymbol{d}\right)\right]\\
=&2\mathbb{E}_{\mathcal{P}\sim\mathcal{E}}\mathbb{E}_{(\boldsymbol{x},\boldsymbol{d})\sim\mathcal{P}}\left[\mathcal{R}_{M}(\mathcal{G})\right],
\end{aligned}
\tag{8}
$$

where σ is a vector of Rademacher random variables. The first inequality comes from moving the supremum inside the expectation, the second inequality is due to the subadditivity of suprema, and the final inequality comes from moving the expectation outside of the supremum. Replacing one of the $(\boldsymbol{x},\boldsymbol{d})$ pairs in the final equality will result in $\mathcal{R}_M(\mathcal{G})$ changing by at most B/M, McDiarmid's inequality can be used to say with confidence $1-\frac{\delta}{2}$ that

$$
2\mathbb{E}_{\mathcal{P}\sim\mathcal{E}}\mathbb{E}_{(\boldsymbol{x},\boldsymbol{d})\sim\mathcal{P}}\left[\mathcal{R}_{M}(\mathcal{G})\right]\leq 2\mathcal{R}_{M}(\mathcal{G})+2\sqrt{\frac{\ln\left(2/\delta\right)}{2M}}. \tag{9}
$$

Combining Eqs. (4), (7) and (9) using the union bound, with confidence at least $1-2\delta$ we have

$$
R^{\varepsilon}\left(h\right)\leq\hat{R}^{\varepsilon}\left(h\right)+2\mathcal{R}_{MN}\left(\mathcal{G}\right)+2\mathcal{R}_{M}\left(\mathcal{G}\right)+3B\sqrt{\frac{\ln\left(2/\delta\right)}{2MN}}+3B\sqrt{\frac{\ln\left(2/\delta\right)}{2M}}. \tag{10}
$$

According to [15], $\ell\left(h\left(\boldsymbol{x}\right),\boldsymbol{d}\right)$ satisfies $\sqrt{2c}$-Lipschitz. According to [13], the following inequalities holds:

$$
\mathcal{R}_{MN}\left(\mathcal{G}\right)\leq 2\sqrt{c}\mathcal{R}_{MN}\left(\mathcal{H}\right), \tag{11}
$$

$$
\mathcal{R}_{M}\left(\mathcal{G}\right)\leq 2\sqrt{c}\mathcal{R}_{M}\left(\mathcal{H}\right). \tag{12}
$$

Plug (11) and (12) into (10), which finishes the proof.

□

A.2 ALGORITHM ADAPTATION METHODS

A.2.1 Domain Shifts with Uncertainty

Domain Shifts with Uncertainty (DSU) [12] is a representative data manipulation method for DG, which characterizes the feature statistics of images as uncertain distributions and hypothesizes that the feature statistics follow a multivariate Gaussian distribution. In DSU, given $\bar{\boldsymbol{x}}\in\mathbb{R}^{B\times C\times H\times W}$ as the encoded features in the intermediate layers of the network, the channel-wise feature mean $\mu\in\mathbb{R}^{B\times C}$ and variance $\sigma^2\in\mathbb{R}^{B\times C}$ of each instance in a mini-batch are formulated as:

$$
\mu\left(\bar{\boldsymbol{x}}\right)=\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\bar{\boldsymbol{x}}_{b,c,h,w}, \tag{13}
$$

$$\sigma^2(\bar{\boldsymbol{x}}) = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(\bar{\boldsymbol{x}}_{b,c,h,w} - \mu(\bar{\boldsymbol{x}})\right)^2. \tag{14}$$

Then, it adopts a non-parametric strategy for uncertainty estimation, utilizing the variance of the feature statistics to provide some instructions:

$$\Sigma_\mu^2(\bar{\boldsymbol{x}}) = \frac{1}{B}\sum_{b=1}^{B}\left(\mu(\bar{\boldsymbol{x}}) - \mathbb{E}_b\left[\mu(\bar{\boldsymbol{x}})\right]\right), \tag{15}$$

$$\Sigma_\sigma^2(\bar{\boldsymbol{x}}) = \frac{1}{B}\sum_{b=1}^{B}\left(\sigma(\bar{\boldsymbol{x}}) - \mathbb{E}_b\left[\sigma(\bar{\boldsymbol{x}})\right]\right), \tag{16}$$

where $\Sigma_\mu^2 \in \mathbb{R}^C$ and $\Sigma_\sigma^2 \in \mathbb{R}^C$ represent the uncertainty estimation of μ and σ, respectively. After that, it leverages the uncertainty estimation captured from the mini-batch to replace the feature statistics with the randomly drawn ones to achieve the transformation using the reparameterization trick [9]:

$$DSU(\bar{\boldsymbol{x}}) = \left(\sigma(\bar{\boldsymbol{x}}) + \varepsilon_\sigma \Sigma_\sigma(\bar{\boldsymbol{x}})\right)\left(\frac{\bar{\boldsymbol{x}} - \mu(\bar{\boldsymbol{x}})}{\sigma(\bar{\boldsymbol{x}})}\right) + \left(\mu(\bar{\boldsymbol{x}}) + \varepsilon_\mu \Sigma_\mu(\bar{\boldsymbol{x}})\right), \tag{17}$$

where $\varepsilon_\mu \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and $\varepsilon_\sigma \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ both follow the standard Gaussian distribution.

To adapt DSU to the GLDL problem, we set the Kullback-Leibler (KL) divergence between the ground truth and the predicted label distribution as the loss function. The objective function can be formulated by

$$\mathcal{L}_{AA-DSU} = \sum_{i=1}^{MN_b}\sum_{j=1}^{c} d_{\boldsymbol{x}_i}^{y_j} \ln \frac{d_{\boldsymbol{x}_i}^{y_j}}{\bar{\boldsymbol{d}}_i^{[j]}}, \tag{18}$$

where MN_b is the number of samples in a batch, $\bar{\boldsymbol{d}}_i^{[j]}$ is the output for the j-th label of $\hat{g}_d\left(g_x\left(f\left(\tilde{\boldsymbol{x}}_i\right)\right)\right)$ and $\tilde{\boldsymbol{x}}_i$ is the augmented instance after applying Eq.(17). Since the DSU does not require interaction between different domains, we indiscriminately sample training instances from each training domain and use them to train the model.

A.2.2 Invariant Risk Minimization

Invariant Risk Minimization (IRM) [1] is a famous representation learning algorithm for the DG problem. It tries to find an invariant data representation from multiple causally related environments. Specifically, it establishes an invariant predictor $\omega \circ \Phi$, where Φ is the data representation, and ω is a classifier simultaneously optimal for all environments. It tries to enforce the optimal classifier on top of the representation space to be the same across all domains. It instantiates the object as:

$$\min_{\Phi:\mathcal{X}\to\mathcal{Y}} \sum_{m=1}^{M} R^{(m)}\left(\Phi(\boldsymbol{x})\right) + \lambda \cdot \left\|\nabla_{\omega|\omega=1.0} R^{(m)}\left(\omega \cdot \Phi(\boldsymbol{x})\right)\right\|, \tag{19}$$

where R denotes the risk function, Φ instantiated as the entire invariant predictor, $\omega = 1.0$ is a scalar and fixed "dummy" classifier and $\lambda \in [0, \infty)$ is a balancing factor.

In GLDL, we instantiate Φ as $h \circ f$, and use the KL divergence as the loss function. The objective function can be formulated by

$$\begin{aligned}\mathcal{L}_{AA-IRM} = &\sum_{m=1}^{M}\sum_{i=1}^{N_b} \mathsf{KL}\left(\boldsymbol{d}_i^{(m)} \middle\| \bar{\boldsymbol{d}}_i^{(m)}\right) \\ &+\lambda \cdot \left\|\nabla_{\omega|\omega=1.0} \mathsf{KL}\left(\boldsymbol{d}_i^{(m)} \middle\| \tilde{\boldsymbol{d}}_i^{(m)}\right)\right\|,\end{aligned} \tag{20}$$

where KL denotes the KL divergence, N_b is the number of samples for each domain in a batch, $\bar{\boldsymbol{d}}_i^{(m)} = \hat{g}_d\left(g_x\left(f\left(\tilde{\boldsymbol{x}}_i^{(m)}\right)\right)\right)$, $\tilde{\boldsymbol{d}}_i^{(m)} = \omega \cdot \hat{g}_d\left(g_x\left(f\left(\tilde{\boldsymbol{x}}_i^{(m)}\right)\right)\right)$ and λ is set to 1.0.

A.2.3 Variance Risk Extrapolation

Variance Risk Extrapolation (VREx) [10] designs a distributionally robust optimization strategy for DG. It reduces the variance of training domain risks to mitigate the sensitivity to a wide range of extreme distributional shifts. Its object can be formulated as:

$$\min_{\theta} \beta \cdot \mathtt{Var}\left(\bar{R}^{(1)}(\theta), \cdots, \bar{R}^{(M)}(\theta)\right) + \sum_{m=1}^{M} \bar{R}^{(m)}(\theta), \tag{21}$$

where $\mathtt{Var}$ denotes the variance function, θ is model parameters, $\bar{R}^{(m)}(\theta)$ is the risk function for the m-th domain and $\beta \in [0, \infty)$ is the balancing factor.

In GLDL, we reshape the object function of VREx as

$$\begin{aligned}\mathcal{L}_{AA-VREx} = & \sum_{m=1}^{|M|} \sum_{i=1}^{N_b} \mathsf{KL}\left(\boldsymbol{d}_i^{(m)} \middle\| \bar{\boldsymbol{d}}_i^{(m)}\right) \\ & + \beta \cdot \mathtt{Var}\left(\sum_{i=1}^{N_b} \mathsf{KL}\left(\boldsymbol{d}_i^{(1)} \middle\| \bar{\boldsymbol{d}}_i^{(1)}\right), \cdots, \sum_{i=1}^{N_b} \mathsf{KL}\left(\boldsymbol{d}_i^{(M)} \middle\| \bar{\boldsymbol{d}}_i^{(M)}\right)\right),\end{aligned} \tag{22}$$

where N_b is the number of samples for each domain in a batch, $\bar{\boldsymbol{d}}_i^{(m)} = \hat{g}_d\left(g_x\left(f\left(\tilde{\boldsymbol{x}}_i^{(m)}\right)\right)\right)$ and β is set to 1.0.

A.3 EVALUATION CRITERIA

Six standard LDL measures (Chebyshev Distance, Clark Distance, Canberra Metric, Kullback-Leibler Divergence, Cosine Coefficient, and Intersection Similarity between ground-truth description degree vectors and predicted description degree vectors) are selected to evaluate different methods for the prediction of the whole distribution. Below we use $\boldsymbol{d}$ to denote the true description degree distribution, $\bar{\boldsymbol{d}}$ to denote the predicted description degree distribution and c to denote the number of labels.

The definition of Chebyshev Distance is:

$$Dis_1\left(\boldsymbol{d}, \bar{\boldsymbol{d}}\right) = \max_j \left|\boldsymbol{d}^{[j]} - \bar{\boldsymbol{d}}^{[j]}\right|.$$

The definition of Clark Distance is:

$$Dis_2\left(\boldsymbol{d}, \bar{\boldsymbol{d}}\right) = \sqrt{\sum_{j=1}^{c} \frac{\left(\boldsymbol{d}^{[j]} - \bar{\boldsymbol{d}}^{[j]}\right)^2}{\left(\boldsymbol{d}^{[j]} + \bar{\boldsymbol{d}}^{[j]}\right)^2}}.$$

The definition of Canberra Metric is:

$$Dis_3\left(\boldsymbol{d}, \bar{\boldsymbol{d}}\right) = \sum_{j=1}^{c} \frac{\left|\boldsymbol{d}^{[j]} - \bar{\boldsymbol{d}}^{[j]}\right|}{\boldsymbol{d}^{[j]} + \bar{\boldsymbol{d}}^{[j]}}.$$

The definition of Kullback-Leibler Divergence is:

$$Dis_4\left(\boldsymbol{d}, \bar{\boldsymbol{d}}\right) = \sum_{j=1}^{c} \boldsymbol{d}^{[j]} \ln \frac{\boldsymbol{d}^{[j]}}{\bar{\boldsymbol{d}}^{[j]}}.$$

The definition of Cosine Coefficient is:

$$Sim_1\left(\boldsymbol{d}, \bar{\boldsymbol{d}}\right) = \frac{\sum_{j=1}^{c} \boldsymbol{d}^{[j]} \bar{\boldsymbol{d}}^{[j]}}{\sqrt{\sum_{j=1}^{c} \boldsymbol{d}^{[j]^2}} \sqrt{\sum_{j=1}^{c} \bar{\boldsymbol{d}}^{[j]^2}}}.$$

The definition of Intersection Similarity is:

$$Sim_2\left(\boldsymbol{d}, \bar{\boldsymbol{d}}\right) = \sum_{j=1}^{c} \min\left(\boldsymbol{d}^{[j]}, \bar{\boldsymbol{d}}^{[j]}\right).$$

A.4 DETAILS OF COMPARED METHODS

- SA-BFGS [4] optimizes the Kullback-Leibler divergence between the ground truth and the predicted distribution through the quasi-Newton method BFGS. In the experiments, the minimum log-likelihood difference for convergence is set to 10^{-5}.
- LALOT [17] casts the label correlations exploration as a ground metric learning problem and adopts optimal transport distance to measure the quality of prediction. In the experiments, the maximum number of iterations is set to 100, λ is set to 0.2, and C is set to 20 as suggested in the code provided by the authors.
- EDL-LRL [7] exploits local label correlation by capturing low-rank structure on clusters of samples with trace-norm regularization. In the experiments, the number of clusters is set to 5 and the maximum number of iterations is set to 50 as suggested in the code provided by the authors.
- LDL-SCL [5] exploits label correlations on local sample. We use its enhancement version that uses the Adam algorithm [8] to optimize the LDL model. In the experiments, the number of clusters is set to 5, the learning rate is set to 0.001, the maximum number of iterations is set to 100, the parameters λ_1, λ_2 and λ_3 are set to 10^{-3} as suggested in the code provided by the authors.
- LDL-LDM [16] exploits both global and local label correlations in a data-driven way. Its basic idea is that label distribution lies in a probability simplex whose underlying structure may encode label correlation. In the experiments, λ_1, λ_2, λ_3 and g are set to 10^{-3}, 1, 1 and 10 respectively as suggested in the code provided by the authors.
- LDL-LRR [6] considers not only the ground-truth label distribution but also the label ranking relation. It adopts a novel ranking loss function for LDL and combines it with K-L divergence as the loss term of the objective function. In the experiments, the parameters λ and β are set to 1, respectively. The initialization of other variables is all-zero, as suggested in the original manuscript.

REFERENCES

[1] Martín Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. 2019. Invariant Risk Minimization. *CoRR* abs/1907.02893 (2019).
[2] Peter L. Bartlett and Shahar Mendelson. 2002. Rademacher and Gaussian Complexities: Risk Bounds and Structural Results. *J. Mach. Learn. Res.* 3 (2002), 463–482.
[3] Sung-Hyuk Cha. 2007. Comprehensive Survey on Distance/Similarity Measures between Probability Density Functions. *International Journal of Mathematical models and Methods in Applied Sciences* 1, 4 (2007), 300–307.
[4] Xin Geng. 2016. Label Distribution Learning. *IEEE Trans. Knowl. Data Eng.* 28, 7 (2016), 1734–1748.
[5] Xiuyi Jia, Zechao Li, Xiang Zheng, Weiwei Li, and Sheng-Jun Huang. 2021. Label Distribution Learning with Label Correlations on Local Samples. *IEEE Trans. Knowl. Data Eng.* 33, 4 (2021), 1619–1631.
[6] Xiuyi Jia, Xiaoxia Shen, Weiwei Li, Yunan Lu, and Jihua Zhu. 2023. Label Distribution Learning by Maintaining Label Ranking Relation. *IEEE Trans. Knowl. Data Eng.* 35, 2 (2023), 1695–1707.
[7] Xiuyi Jia, Xiang Zheng, Weiwei Li, Changqing Zhang, and Zechao Li. 2019. Facial Emotion Distribution Learning by Exploiting Low-Rank Label Correlations Locally. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 9841–9850.
[8] Diederik P. Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *Proceedings of the 3rd International Conference on Learning Representations*.
[9] Diederik P. Kingma and Max Welling. 2014. Auto-Encoding Variational Bayes. In *Proceedings of the 2nd International Conference on Learning Representations*.
[10] David Krueger, Ethan Caballero, Jörn-Henrik Jacobsen, Amy Zhang, Jonathan Binas, Dinghuai Zhang, Rémi Le Priol, and Aaron C. Courville. 2021. Out-of-Distribution Generalization via Risk Extrapolation (REx). In *Proceedings of the 38th International Conference on Machine Learning*, Vol. 139. 5815–5826.
[11] Da Li, Henry Gouk, and Timothy M. Hospedales. 2022. Finding lost DG: Explaining domain generalization via model complexity. *CoRR* abs/2202.00563 (2022).
[12] Xiaotong Li, Yongxing Dai, Yixiao Ge, Jun Liu, Ying Shan, and Lingyu Duan. 2022. Uncertainty Modeling for Out-of-Distribution Generalization. In *Proceedings of the 10th International Conference on Learning Representations*.
[13] Andreas Maurer. 2016. A vector-contraction inequality for rademacher complexities. In *Proceedings of the 27th International Conference on Algorithmic Learning Theory*. 3–17.
[14] Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar. 2018. *Foundations of machine learning*. MIT press.
[15] Jing Wang and Xin Geng. 2019. Theoretical Analysis of Label Distribution Learning. In *Proceedings of the 33rd AAAI Conference on Artificial Intelligence*. 5256–5263.
[16] Jing Wang and Xin Geng. 2023. Label Distribution Learning by Exploiting Label Distribution Manifold. *IEEE Trans. Neural Networks Learn. Syst.* 34, 2 (2023), 839–852.
[17] Peng Zhao and Zhi-Hua Zhou. 2018. Label Distribution Learning by Optimal Transport. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*. 4506–4513.